CYCLICITY WHEN THE OMITTED CURVE HAS UNBOUNDED ARGUMENT

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AI assistance was used in adapting and checking the argument and the exposition.

Dan and Guo ask: "In Theorem 7.1, can the assumption that γ has a bounded argument function be dropped?" See Question 7.6, p. 42 of their paper on ordinary Dirichlet series and dilation systems. The theorem concerns a zero-free $F \in H^2(\mathbb{T}^\infty)$ whose range on $X = \mathbb{D}^\infty \cap \ell^2$ avoids a simple curve from zero to infinity.

Theorem 7.1 asserts cyclicity of a zero-free $F \in H^2(\mathbb{D}_2^\infty)$ whose range avoids a simple curve γ joining zero to infinity, provided γ has a bounded continuous argument; $\mathbb{D}_2^\infty = \mathbb{D}^\infty \cap \ell^2$, and cyclicity means that polynomial multiples of F are dense.

Dan and Guo, *The Periodic Dilation Completeness Problem*, arXiv:1908.03113v2.

Answered: yes. Evidence: Sketch.

Yes: every such F is cyclic, without a bounded-argument assumption on the omitted curve.

Proof. We first show that if the range of a zero-free $F \in H^2(\mathbb{T}^\infty)$ on X lies in a simply connected domain $\Omega \subset \mathbb{C} \setminus \{0\}$, then F is cyclic. Let $\psi : \mathbb{D} \to \Omega$ be conformal and write $F = \psi \circ h$, where $h = \psi^{-1} \circ F$ maps X to $\mathbb{D}$. A univalent function omitting zero and its reciprocal belong to $H^q(\mathbb{D})$ for every $q < 1/2$. Consequently $u = \log|\psi^*| \in L^2(\mathbb{T})$ and the outer factorization has the form $\psi = e^{i\alpha}\exp(Hu)$, where H is the analytic Poisson extension. Truncate u from below by $u_\varepsilon = \max(u, \log\varepsilon)$ and set $b_\varepsilon = e^{-i\alpha}\exp(-Hu_\varepsilon)$. Then b_ε is bounded analytic, $\|b_\varepsilon\|_\infty \le \varepsilon^{-1}$, $|\psi b_\varepsilon| \le 1$, and $\psi b_\varepsilon \to 1$ locally uniformly. The bounded analytic multiplier $b_\varepsilon \circ h$ shows $A_\varepsilon = (\psi b_\varepsilon) \circ h$ belongs to the cyclic subspace generated by F. Also $\|A_\varepsilon\|_2 \le 1$ and $A_\varepsilon(0) \to 1$, so $\|A_\varepsilon - 1\|_2^2 \le 2 - 2\operatorname{Re} A_\varepsilon(0) \to 0$. Thus 1 is in that subspace.

For a simple curve γ joining 0 to infinity, its closure in the Riemann sphere is a Jordan arc. Its complement is a simply connected domain avoiding zero. The image $F(X)$ is connected and avoids γ, so it lies in this domain. The preceding argument applies and proves cyclicity. This removes only the bounded-argument restriction; zero-freeness and the range-avoidance hypothesis remain. □

Date: 13 October 2026.